

# Homework 1: Discrete State Dynamic Programming

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## 1 Overview

I solve the two-state Huggett household problem at the fixed prices implied by  $K^0 = 5$ . The final numerical specification uses modified Howard improvement, sparse CSR power iteration, an exp-log asset grid on  $[0, 5]$ , and  $N = 5000$ . The Bellman formulation and Euler derivation are given in `manual/derivation.pdf`; generated tables and figures are included from `results/`.

## 2 Solver Comparison: Part (a)

The baseline uses a uniform  $[0, 20]$  grid with  $N = 1000$  and zero initial values. Convergence requires

$$d(x) = \max_m \frac{|T(x)_m - x_m|}{|x_m| + 1} \leq 10^{-8}.$$

Pass counts include the terminating check, and solver times exclude the common utility-matrix construction.

Table 1: Baseline solver comparison,  $N = 1000$  on  $[0, 20]$ .

| Method          | Passes | Updates | Inner steps | Seconds   | Exit metric       | Status             |
|-----------------|--------|---------|-------------|-----------|-------------------|--------------------|
| vfi             | 379    | 378     | 0           | 0.923665  | 9.71806129077e-09 | converged          |
| howard          | 15     | 14      | 0           | 0.063859  | 6.01925131926e-16 | converged          |
| modified_howard | 19     | 18      | 360         | 0.055136  | 9.33987779834e-09 | converged          |
| gradient        | 20000  | 20000   | 0           | 51.552503 | 1.25760126001     | capped_unconverged |
| adam            | 20000  | 20000   | 0           | 51.727502 | 0.125113107262    | capped_unconverged |

VFI, Howard, and modified Howard converge to identical policies in 379, 15, and 19 passes. Modified Howard is the fastest converged method at 0.055136 seconds, so I select it. The contraction benchmark  $\log(10^{-8})/\log(0.96) = 451.244$  is larger than the observed VFI count because it is a contraction-based error-reduction benchmark, not the exact stopping count for the relative Bellman metric.

The fixed gradient step  $\eta = 0.0005$  was preregistered as a conservative choice: at the baseline size, the bound on  $\|I - \beta Q\|_2^2$  is below 2000, so the step uses the corresponding inverse scale. Fixed-step gradient descent and Adam both hit the 20,000-pass cap and remain unconverged. The requested benchmark  $(1 - \beta)^{-2} = 625$  is reported separately from the measured Euclidean condition numbers. In this problem, the measured  $\text{cond}_2(J^T J)$  is much larger, so 625 should not be read as the measured 2-norm condition number of this nonnormal Jacobian.

Table 2: Contraction and conditioning diagnostics,  $J = I - \beta Q_G$ .

| Quantity                             | Value            |
|--------------------------------------|------------------|
| Contraction expression               | 451.24401589     |
| Observed VFI outer passes            | 379              |
| Specified $1/(1-\beta)^2$ expression | 625              |
| zero_values: sigma_min               | 0.00131755850852 |
| zero_values: sigma_max               | 30.3591830961    |
| zero_values: condition_J             | 23041.9999566    |
| zero_values: condition_JtJ           | 530933762        |
| howard_solution: sigma_min           | 0.0199827991366  |
| howard_solution: sigma_max           | 2.03095066516    |
| howard_solution: condition_J         | 101.634943697    |
| howard_solution: condition_JtJ       | 10329.6617804    |

### 3 Invariant Distribution: Parts (b)–(d)

Using the converged Howard policy, I compare power iteration, the eigenvector method, and direct solution of  $Q_G^T \pi = \pi$  with both dense and CSR representations.

Table 3: Invariant-distribution methods for the common Howard policy.

| Method        | Seconds  | Power updates | Raw minimum        | Correction mass   | Stationarity      |
|---------------|----------|---------------|--------------------|-------------------|-------------------|
| dense_power   | 0.238683 | 212           | 0                  | -0                | 7.76899378163e-13 |
| dense_eigen   | 0.074303 | n/a           | -8.06313061579e-28 | 4.45487966523e-26 | 1.73472347598e-17 |
| dense_direct  | 0.036680 | n/a           | 0                  | -0                | 3.1918911958e-16  |
| sparse_power  | 0.003668 | 212           | 0                  | -0                | 7.76906317057e-13 |
| sparse_eigen  | 0.005532 | n/a           | -4.0315653079e-28  | 2.01578265395e-26 | 1.38777878078e-17 |
| sparse_direct | 0.005262 | n/a           | 0                  | -0                | 8.48773347979e-18 |

Table 4: Sup-norm agreement among invariant distributions.

| Comparison             | Maximum difference | Pair                       |
|------------------------|--------------------|----------------------------|
| dense                  | 5.49010836792e-12  | dense_power, dense_direct  |
| sparse                 | 5.48994877336e-12  | sparse_power, sparse_eigen |
| across_representations | 5.49011530682e-12  | dense_direct, sparse_power |
| all_six                | 5.49011530682e-12  | dense_direct, sparse_power |

All six solutions pass the registered checks, with maximum disagreement about  $5.49 \times 10^{-12}$ . Sparse power iteration is the fastest measured sparse method at 0.003668 seconds, so I select power iteration with CSR for the remaining experiments.

### 4 Asset Range: Part (e)

At  $N = 1000$ , I test uniform grids with  $k_{\max} = 1, 2, 5, 10, 20, 40$ .

The ranges ending at 1 and 2 are rejected because they place positive mass at the top node. I select  $k_{\max} = 5$ , the smallest tested range with zero top-node mass and an upper support endpoint well below the grid boundary.

Table 5: Uniform asset-range comparison at  $N = 1000$ .

| k_max | Top mass          | Support endpoint | Support share | Compute seconds | Range status       |
|-------|-------------------|------------------|---------------|-----------------|--------------------|
| 1     | 0.0237509414545   | 1                | 1.000000      | 0.085518        | rejected_truncated |
| 2     | 6.01277209443e-05 | 2                | 1.000000      | 0.082441        | rejected_truncated |
| 5     | 0                 | 3.21821821822    | 0.634000      | 0.074443        | eligible           |
| 10    | 0                 | 3.21321321321    | 0.322000      | 0.072260        | eligible           |
| 20    | 0                 | 3.26326326326    | 0.164000      | 0.072538        | eligible           |
| 40    | 0                 | 3.24324324324    | 0.082000      | 0.074047        | eligible           |

## 5 Grid Construction: Part (f)

At  $N = 1000$  on  $[0, 5]$ , I compare the uniform grid with

$$k_n = \exp\left[\frac{n-1}{N-1} \log(1 + k_{\max})\right] - 1.$$

Table 6: Uniform and exp-log grids,  $N = 1000$  on  $[0, 5]$ .

| Quantity                  | Uniform           | Exp-log           |
|---------------------------|-------------------|-------------------|
| Outer passes              | 19                | 19                |
| Updates                   | 18                | 18                |
| Inner steps               | 360               | 360               |
| Exit metric               | 8.07103857078e-09 | 8.0834666578e-09  |
| Solver seconds            | 0.0566389579981   | 0.0530802919966   |
| Power updates             | 172               | 168               |
| Distribution seconds      | 0.00352920899604  | 0.00297812499775  |
| Compute seconds           | 0.0831304579915   | 0.0757789579875   |
| Maximum grid step         | 0.00500500500501  | 0.0107516734031   |
| Mean assets               | 0.450551165217    | 0.450853905799    |
| Top mass                  | 0                 | 0                 |
| Support endpoint          | 3.21821821822     | 3.21407251709     |
| Support share             | 0.634             | 0.799             |
| Slack count               | 1978              | 1945              |
| Slack mass                | 0.90977107721     | 0.911966346902    |
| Upper choices             | 0                 | 0                 |
| Euler maximum             | 0.00412096375367  | 0.00612421604424  |
| Euler mean                | 0.00120403136135  | 0.00108912237111  |
| Conditional weighted mean | 0.00147199905766  | 0.0008046570183   |
| Weighted maximum: max(qE) | 0.000102998957978 | 6.13602452215e-05 |
| Supported slack maximum   | 0.00412096375367  | 0.00407219098769  |

Both grids pass all applicable checks. I select the exp-log grid because it has lower mean and distribution-weighted Euler errors, although its unweighted maximum residual is slightly higher.

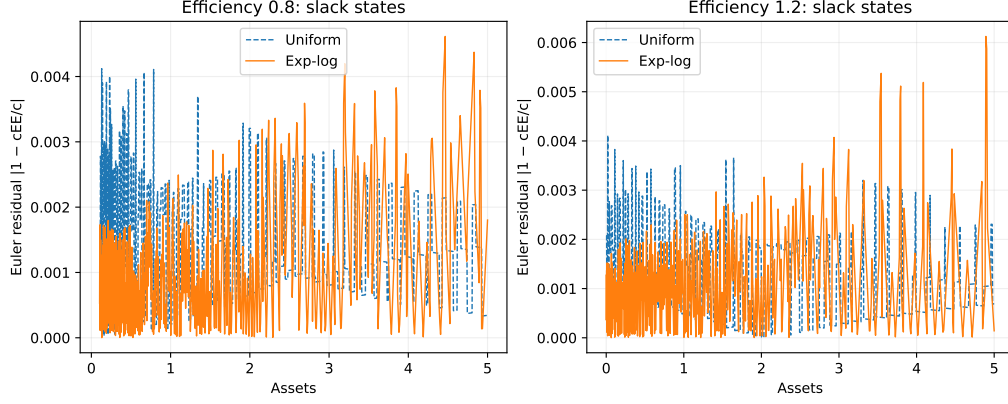


Figure 1: Euler-residual comparison for the uniform and exp-log grids.

## 6 Grid Size: Part (g)

I compare  $N = 100, 500, 1000, 2000, 5000$  under the selected configuration.

Table 7: Grid-size comparison on the exp-log  $[0, 5]$  grid.

| Quantity                  | N=100             | N=500             | N=1000            | N=2000            | N=5000            |
|---------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Outer passes              | 19                | 19                | 19                | 19                | 19                |
| Updates                   | 18                | 18                | 18                | 18                | 18                |
| Inner steps               | 360               | 360               | 360               | 360               | 360               |
| Exit metric               | 8.03364122753e-09 | 8.07799018063e-09 | 8.0834666578e-09  | 8.07986477462e-09 | 8.08314692909e-09 |
| Utility seconds           | 0.000309292001475 | 0.00679545800085  | 0.0241041250047   | 0.112672458003    | 0.655352125003    |
| Solver seconds            | 0.00388824900438  | 0.0156604590011   | 0.0537732079974   | 0.212434167006    | 1.285760583       |
| CSR construction seconds  | 0.000102291996882 | 0.000170540995896 | 0.000141750002513 | 0.000179458002094 | 0.000298332997772 |
| Power seconds             | 0.00271195799724  | 0.00287229099922  | 0.00301779100118  | 0.00373687600222  | 0.00624258299649  |
| Power updates             | 168               | 169               | 168               | 168               | 168               |
| Compute seconds           | 0.00701179099997  | 0.0254987489971   | 0.0810368740058   | 0.329022959013    | 1.947653624       |
| Stationarity residual     | 7.87911402789e-13 | 7.53008766452e-13 | 8.48758563432e-13 | 8.50611248104e-13 | 8.52012904673e-13 |
| Maximum grid step         | 0.107614708755    | 0.0215055689019   | 0.0107516734031   | 0.00537555790023  | 0.00215015611502  |
| Mean assets               | 0.439732993184    | 0.45019078318     | 0.450853905799    | 0.450965370158    | 0.451126535212    |
| Top mass                  | 2.67276471009e-53 | 0                 | 0                 | 0                 | 0                 |
| Support endpoint          | 3.02929208117     | 3.22015069022     | 3.21407251709     | 3.2072684858      | 3.17018140386     |
| Support share             | 0.78              | 0.802             | 0.799             | 0.793             | 0.7888            |
| Slack count               | 193               | 972               | 1945              | 3891              | 9730              |
| Slack mass                | 0.896989880687    | 0.911024525526    | 0.911966346902    | 0.912035439329    | 0.912256038332    |
| Upper choices             | 1                 | 0                 | 0                 | 0                 | 0                 |
| Euler maximum             | 0.0469940723466   | 0.0102519768383   | 0.00612421604424  | 0.00293533782662  | 0.00118455627353  |
| Euler mean                | 0.0117693950252   | 0.00225801889311  | 0.00108912237111  | 0.000563182483879 | 0.000223018069011 |
| Conditional weighted mean | 0.0087004085652   | 0.00150289123749  | 0.0008046570183   | 0.000395872498082 | 0.000155907135197 |
| Weighted maximum: max(qE) | 0.000767549255638 | 0.000121869663975 | 6.13602452215e-05 | 1.72349708195e-05 | 6.44444556925e-06 |
| Supported slack maximum   | 0.0469940723466   | 0.00780556323862  | 0.00407219098769  | 0.00197919372155  | 0.000966044811544 |

All five solutions converge in 19 passes. Euler errors fall steadily as  $N$  increases, while the  $N = 5000$  run remains computationally modest at about 1.95 seconds. I therefore select  $N = 5000$  because it gives the lowest tested Euler errors at an acceptable runtime.

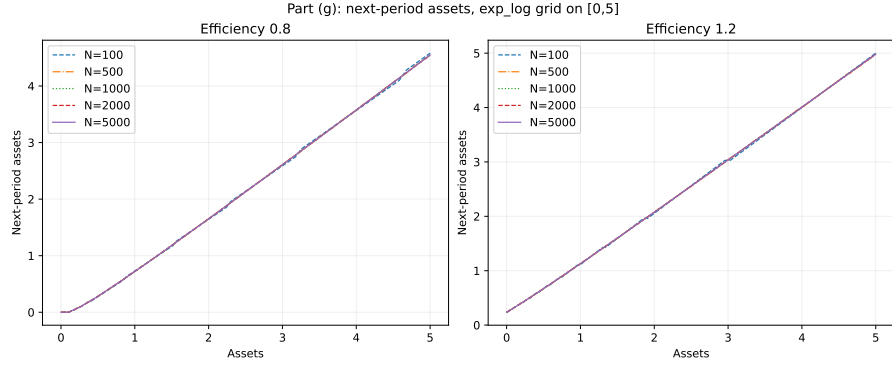


Figure 2: Asset policies across grid sizes.

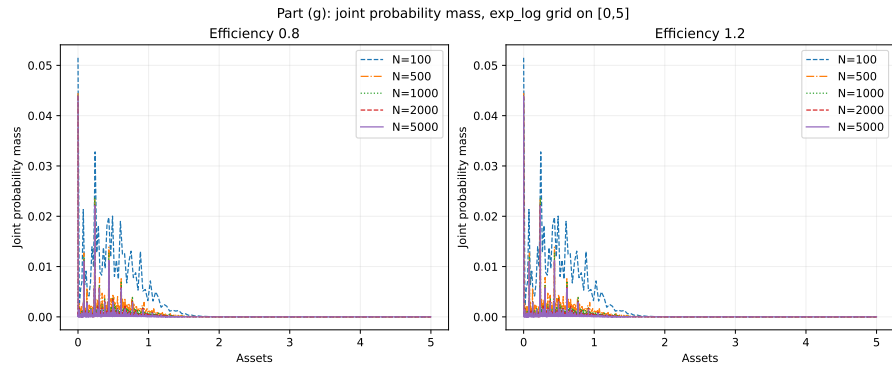


Figure 3: Joint invariant probability masses across grid sizes.

## 7 Euler Accuracy: Part (h)

For slack borrowing states, the consumption-equivalent residual is

$$c_{n,s}^{EE} = \left[ \beta(1+r^0) \sum_{s'} P_{ss'} (c_{n,s,s'}^{\text{next}})^{-\sigma} \right]^{-1/\sigma}, \quad E_{n,s} = \left| 1 - \frac{c_{n,s}^{EE}}{c_{n,s}} \right|.$$

For the slack-state set

$$\mathcal{A} = \{(n, s) : G(n, s) > 0\}, \quad q_{n,s} = \frac{\pi_{n,s}}{\sum_{(i,j) \in \mathcal{A}} \pi_{i,j}}.$$

The weighted maximum is

$$\max_{(n,s) \in \mathcal{A}} q_{n,s} E_{n,s},$$

while the supported slack maximum is

$$\max_{\substack{(n,s) \in \mathcal{A} \\ \pi_{n,s} > 10^{-12}}} E_{n,s}.$$

Table 8: Euler residuals at the selected  $N = 5000$  grid.

| Quantity           | Value             |
|--------------------|-------------------|
| slack_count        | 9730              |
| slack_mass         | 0.912256038332    |
| upper_choice_count | 0                 |
| maximum            | 0.00118455627353  |
| mean               | 0.000223018069011 |
| weighted_mean      | 0.000155907135197 |
| weighted_maximum   | 6.44444556925e-06 |
| supported_maximum  | 0.000966044811544 |

The final mean, maximum, and conditional weighted mean residuals are  $2.23018 \times 10^{-4}$ ,  $1.18456 \times 10^{-3}$ , and  $1.55907 \times 10^{-4}$ . No selected policy chooses the upper grid bound.

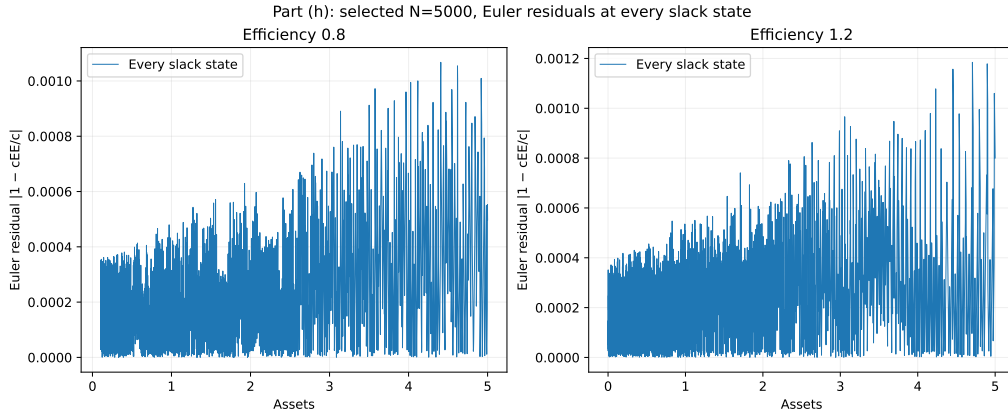


Figure 4: Euler residuals at the selected grid size.

Table 9: Log-log Euler-error scaling across five grids.

| Error summary | Intercept       | Slope          | Used grids | Reason    |
|---------------|-----------------|----------------|------------|-----------|
| mean          | -2.19734219457  | 1.01328727392  | 5          | available |
| maximum       | -0.957782213685 | 0.933748380727 | 5          | available |

Across the five saved grid sizes, the fitted log-log slopes are 1.013287 for the mean error and 0.933748 for the maximum error.

These slopes are close to one, which is consistent with approximately first-order error reduction over the tested grid resolutions.

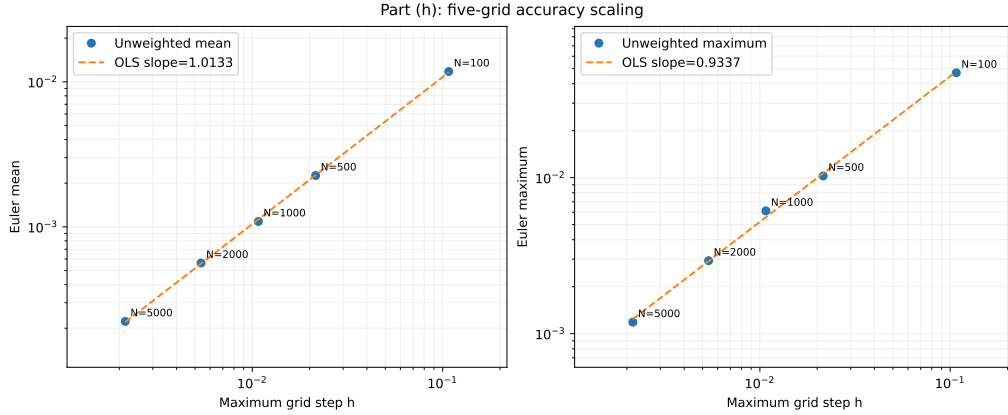


Figure 5: Mean and maximum Euler errors against the maximum grid step.

**Final checks.** All nine selected-grid checks pass. A total of 202 check records pass, including the 30 Stage-3 validation records, with two expected rejected trial ranges. The reproduction check matches 330 saved arrays and 8,514 JSON numerical values with zero numerical difference; the capped gradient and Adam runs remain correctly labeled unconverged.

## 8 Independent Verification

**Manual-kernel check.** On October 1, 2026, I independently reran the manual-kernel comparison on the  $N = 100$  uniform  $[0, 20]$  grid. Both the manual and implementation VFI runs required 492 passes. The maximum value-function difference was  $3.71 \times 10^{-9}$  and there were zero policy mismatches, satisfying the  $10^{-8}$  comparison criterion.

**Final Euler check.** On October 1, 2026, I also independently checked the final  $N = 5000$  Euler diagnostics. The mean and maximum residuals were 0.000223018 and 0.001184556, matching the saved Stage-4 results to numerical precision.

These checks are separate from the execution agent’s automated Stage-3 and Stage-4 verification and are recorded in `log.md`.

## 9 AI-Use Note

I used Codex (GPT-6; exact build unavailable) through a desktop/local-folder agent workflow to implement the production solvers and tests, run the numerical experiments, and generate the results, tables, figures, and reproduction evidence. I also used ChatGPT text chat to help draft and revise student-owned workflow files, later manual-kernel maintenance edits, and this report draft. I made the recorded runtime choices and reasons, reviewed the evidence, recorded the verification in `log.md`, and am responsible for the final report and Stage-5 commit. The report-check agent may correct only numbers, references, and typographical errors.

## 10 Conclusion

Modified Howard with CSR power iteration on the  $N = 5000$  exp-log  $[0, 5]$  grid passes the registered checks and yields small Euler residuals with approximately first-order scaling over the tested resolutions.